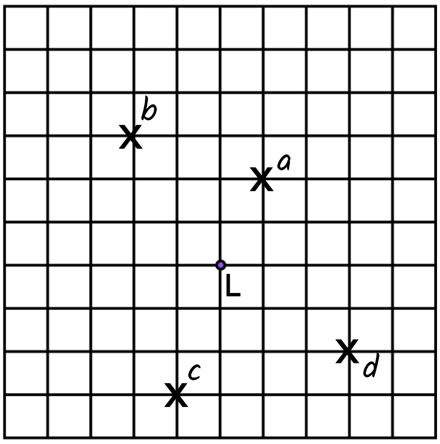
Translating objects

Task A: Translating a single point from a description

1) Mark the image of point L after the following translations:

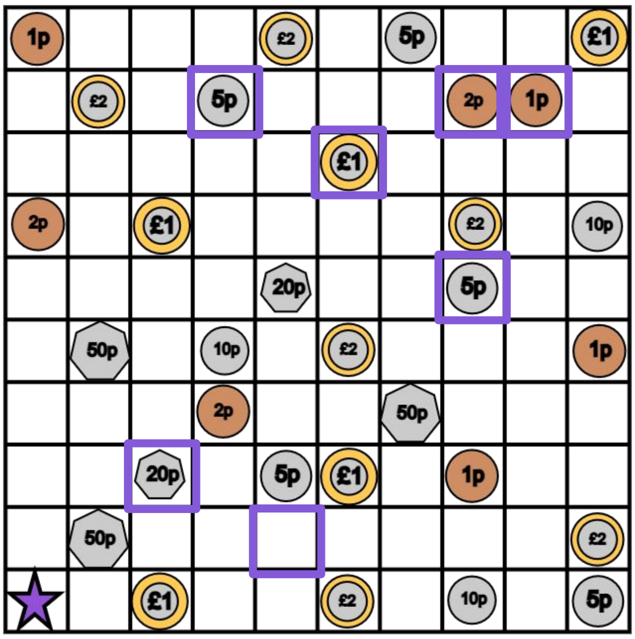
a) $\begin{pmatrix}1\\2\end{pmatrix}$ b) $\begin{pmatrix}-2\\3\end{pmatrix}$ c) $\begin{pmatrix}-1\\-3\end{pmatrix}$ d) $\begin{pmatrix}3\\-2\end{pmatrix}$



2) Work out the total money collected when following this vector trail.
Start on the star.

$$\begin{pmatrix}2\\2\end{pmatrix} \to \begin{pmatrix}2\\-1\end{pmatrix} \to \begin{pmatrix}4\\7\end{pmatrix} \to \begin{pmatrix}-3\\-1\end{pmatrix} \to \begin{pmatrix}-2\\1\end{pmatrix} \to \begin{pmatrix}4\\0\end{pmatrix} \to \begin{pmatrix}0\\-3\end{pmatrix}$$

£1.33



3) Investigate the total money you can collect using all of the vectors but in any order.
Start on the star.

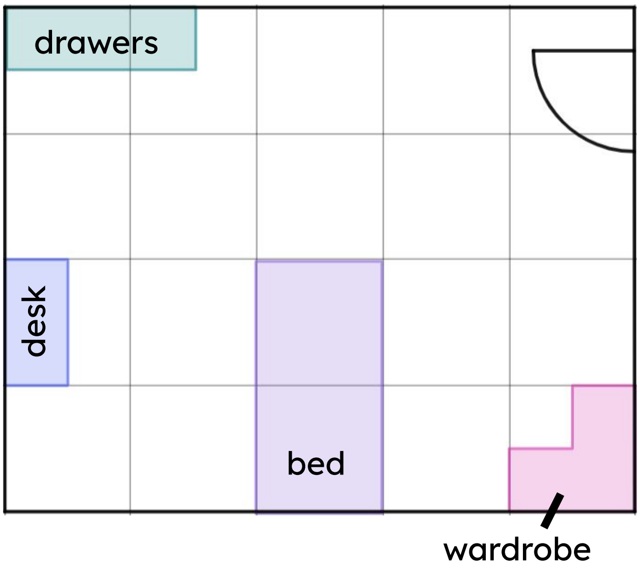
A= $\begin{pmatrix}2\\2\end{pmatrix}$ B= $\begin{pmatrix}2\\-1\end{pmatrix}$ C= $\begin{pmatrix}4\\7\end{pmatrix}$ D= $\begin{pmatrix}-2\\1\end{pmatrix}$ E= $\begin{pmatrix}4\\0\end{pmatrix}$

F= $\begin{pmatrix}-3\\-1\end{pmatrix}$ G= $\begin{pmatrix}0\\-3\end{pmatrix}$

Some examples:
A-C-F-B-G-E-D £3.36
E-A-D-F-C-G-B 5p

Task B: Translating more complex objects

1) Here is a plan view of a bedroom. By only translating the objects, redesign the layout.



Here is an example.

The pink corner wardrobe cannot move anywhere else, if it is to remain touching two walls.

2) Complete the translations, always start with triangle A.

a) $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ b) $\begin{pmatrix} -3 \\ -2 \end{pmatrix}$

c) $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ d) $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$

e) $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ f) $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$